

INTRODUZIONE:

PARTICELLE $\Rightarrow$ VARI

legge di Newton

- massa m
- posiz. $\vec{r}$
- Traiettoria $\vec{r} = \vec{r}(t)$
- Velocità $\vec{v} = \frac{d\vec{r}}{dt}$
- quantità di moto $\vec{p} = m\vec{v}$
(momento lineare)
- $K = \frac{1}{2} m v^2$ (localizzato)

Energia tot. particella

$$E^2 = p^2 c^2 + m^2 c^4$$

$$E = \sqrt{p^2 c^2 + m^2 c^4} = \sqrt{m^2 c^4 \left(\frac{p^2}{m^2 c^2} + 1 \right)} = \left\{ \begin{array}{l} p \ll mc \\ \downarrow \\ v \ll c \end{array} \right\} \approx mc^2 \left(1 + \frac{p^2}{2m^2 c^2} \right) = \overbrace{mc^2}^{E_0 \text{ (costante)}} + \frac{p^2}{2m} \underset{v E_c}{\approx}$$

$\rightarrow$ regime non relativistico

$$\Rightarrow \text{REGIME NON RELAT} \Rightarrow E = \frac{p^2}{2m} = \frac{1}{2} m v^2$$

• QUANTIZZAZIONE della materia: Lucrezio, Democrito (V sec. a.C.)

$$\approx 100 \Rightarrow \text{Atomi} \Rightarrow \text{Tabella periodica} \Rightarrow \text{Mole} \Rightarrow N_A = 6,02 \cdot 10^{23}$$

• " della carica: $q_e = -1,6 \cdot 10^{-19} \text{ C}$

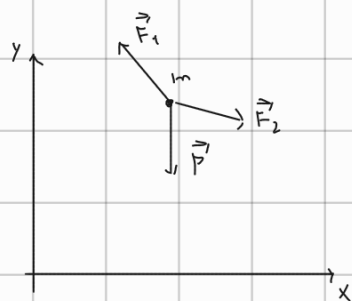
• " dell' ENERGIA: Planck

$$E = n(h\nu) \quad \begin{array}{l} \uparrow 1, 2, 3 \dots \\ \downarrow \text{Frequenza} \\ \downarrow \text{costante di Planck} \\ h = 6,626 \cdot 10^{-34} \text{ Js} \end{array}$$

MOMENTO ANGOLARE

$$E = n(h\omega) \quad \downarrow \quad h = \frac{h}{2\pi} \text{ costante di Planck ridotta}$$

• FORMALISMO LAGRANGIANO:



$$F_x^{\text{tot}} = m a_x \Rightarrow m \frac{d^2 x(t)}{dt^2} = F_x^{\text{tot}}$$

$$\left\{ \begin{array}{l} x(0) \\ \frac{dx}{dt}(0) \end{array} \right\} \xrightarrow{\text{Legge del mot.}} x = x(t)$$

GdL: n di param. indipendenti per descrivere intern. il sistema

n GdL $\Rightarrow$ n coord. generalizzate $q_1, q_2, q_3, \dots, q_n \Rightarrow \dot{q}_i = \frac{dq_i}{dt}$ vel. generaliz.

Scopo: $q = q(t)$

LAGRANGIANA:

$$\mathcal{L} = \mathcal{L}(q_1, \dots, q_n; \dot{q}_1, \dots, \dot{q}_n, t) = \underbrace{K(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)}_{\text{Energ. cinet.}} - \underbrace{V(q_1, \dots, q_n, t)}_{\text{Energ. pot.}}$$

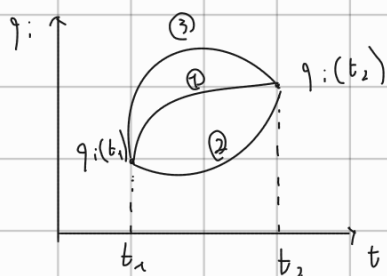
$$= K(q, \dot{q}) - V(q, t) \rightarrow \text{Sistemi conservativi}$$

$\rightarrow$ Sistema di N punti mot. $K = \sum_{i=1}^N \frac{1}{2} m_i |\vec{v}_i|^2$ $V = V(\vec{r}_1, \dots, \vec{r}_N, t)$

$\downarrow$
 $\dot{q}_{ix}, \dot{q}_{iy}, \dot{q}_{iz}$

Azione

$$S = \int_{t_1}^{t_2} \mathcal{L}(q(t), \dot{q}(t), t) dt \quad [S] = E \cdot t \Rightarrow \text{Momento angolare}$$



FAMIGLIA di possibili $q(t)$

$$S = \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}, t) dt \rightarrow \text{FUNZIONALE (la } q \text{ non è univocamente definita)}$$

$\rightarrow$ PRINCIPIO di HAMILTON (Azione stazionaria):

$$\delta S = \int_{t_1}^{t_2} \mathcal{L}(q_i, \dot{q}_i, t) dt = 0$$

 q tale per cui S è stazionaria

$\hookrightarrow$ differenziale di un funzionale

Es.:

$$\mathcal{L} = K(q, \dot{q}) - V(q)$$

$$\delta S = \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}) dt = 0 \Rightarrow \delta S = \int_{t_1}^{t_2} \left(\frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial \dot{q}} \delta \dot{q} \right) dt =$$

$\downarrow$
diff. tra due $q(t)$
infinitamente vicine

$$= \int_{t_1}^{t_2} \left(\frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial \dot{q}} \frac{d}{dt} (\delta q) \right) dt = 0$$

$\downarrow$

$$\int_{t_1}^{t_2} \left[\underbrace{\frac{\partial \mathcal{L}}{\partial \dot{q}}}_{\mathcal{L}(t)} \underbrace{\frac{d}{dt}(\dot{q})}_{\dot{q}'(t)} \right] dt = \cancel{\frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{q} \Big|_{t_1}^{t_2}} - \int_{t_1}^{t_2} \underbrace{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right)}_{\dot{q}(t)} \dot{q} dt$$

$\downarrow$
 $\dot{q}(t_1) = \dot{q}(t_2) = 0$

$$\Rightarrow \int_{t_1}^{t_2} \left(\frac{\partial \mathcal{L}}{\partial q} \dot{q} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \dot{q} \right) dt = 0$$

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \right] \underbrace{\int_{t_1}^{t_2} \dot{q} dt}_{\text{arbitrario}} = 0 \Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial t} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) = 0}$$

$= 0$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = \frac{\partial \mathcal{L}}{\partial q_i} \quad \text{Eq. di Lagrange}$$

Momento canonico coniugato alla coord. q_k

$$p_k = \frac{\partial \mathcal{L}}{\partial \dot{q}_k} \Rightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) = \frac{\partial \mathcal{L}}{\partial q_k}$$

ANALOGO della legge di NEWTON

$$\mathcal{L} \neq \mathcal{L}(q_k) \Rightarrow \frac{\partial \mathcal{L}}{\partial q_k} = 0$$

$\Downarrow$

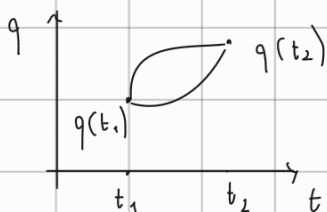
$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) = 0 = \frac{d}{dt} (p_k) = 0 \Rightarrow p_k = \text{cost}_j$$

$$\Rightarrow q_k \text{ è ciclica}$$

① EQ. SCALARI (non ho vettori)

② q_k (ciclica) $\Rightarrow p_k = \text{cost}$

③ $x = x(t) \quad m \frac{d^2 x(t)}{dt^2} = F_x^{\text{tot}}$



HAMILTON è integrale $\Rightarrow$ considera già tutte le traiettorie

non punto per punto. considera tutti i percorsi e sceglie quello che rende minima l'azione

Es.: ① Partic. nel campo 3D

$$q := (x, y, z) \quad V = mgz \quad \mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz \quad (3 \text{ variabili})$$

$\hookrightarrow 3$ eq. di Lag.

$$i = x \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x} \Rightarrow \frac{d}{dt} \underbrace{(m \dot{x})}_p = 0 \quad \text{uguale per } i = y$$

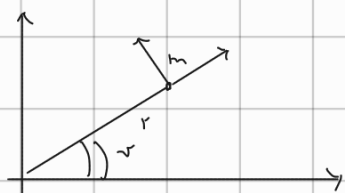
$$\Rightarrow p_x = \text{cost}$$

$$i = z \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{z}} \right) = \frac{\partial \mathcal{L}}{\partial z} \Rightarrow \frac{d}{dt} (m \dot{z}) = -mg$$

$$\Downarrow$$

$$m \underbrace{\frac{d^2 z}{dt^2}}_{\ddot{z}} + mg = 0 \quad (\text{NEWTON})$$

Es. ② Part. in un piano con $F = F(r)$



$$\vec{r} = r \vec{u}_r + r \dot{\phi} \vec{u}_\phi$$

$$\vec{v} = \dot{r} \vec{u}_r + r \ddot{\phi} \vec{u}_\phi + r \dot{\phi} \dot{\phi} \vec{u}_\phi$$

$$K = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\phi}^2) \quad V = V(r)$$

$$\mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\phi}^2) - V(r)$$

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = m r^2 \dot{\phi} = m \omega r^2 = \underbrace{|\vec{L}|}_{\text{mom. Ang}} \quad (\text{SI CONSERVA})$$

FORMALISMO HAMILTONIANO:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = \frac{\partial \mathcal{L}}{\partial q_i} \quad i = 1, \dots, n \quad \text{in Eq. II° ordine}$$

$$\begin{cases} q_i(0) \\ \dot{q}_i(0) = \frac{dq_i}{dt}(0) \end{cases}$$

$$\underbrace{(q_i, \dot{q}_i)}_{\text{non indipendenti}} \Leftrightarrow \underbrace{(q_i, p_i)}_{\text{indipendenti}} \Rightarrow H(q, p) = \sum_{i=1}^n p_i \dot{q}_i - \mathcal{L}(q, \dot{q})$$

indipendenti
n variabili

2n variabili

$$\rightarrow \text{Se sistema conserv.} \Rightarrow H(q, p) = K(q, p) + V(q)$$

$$\begin{aligned} q_i(0) &\Rightarrow \begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ p_i(0) \Rightarrow \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{cases} \quad i = 1, \dots, n \end{aligned}$$

Eq. di HAMILTON

(ho raddopp. le var ma ora le eq. sono del I ordine)

Es.:

$H(q, p)$ per particella con $V(x, y, z)$

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - V(x, y, z) \Rightarrow H = \sum_{i=1}^3 p_i \dot{q}_i - \mathcal{L} = \\ &= p_x \dot{x} + p_y \dot{y} + p_z \dot{z} - \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + V(x, y, z) \quad \uparrow \quad p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \Rightarrow p_x = m \dot{x} \Rightarrow \dot{x} = \frac{p_x}{m} \\ &= \frac{p_x^2}{m} + \frac{p_y^2}{m} + \frac{p_z^2}{m} - \frac{1}{2} m \left(\frac{p_x^2}{m^2} + \frac{p_y^2}{m^2} + \frac{p_z^2}{m^2} \right) + V(x, y, z) = \\ &= \underbrace{\frac{1}{2m} (p_x^2 + p_y^2 + p_z^2)}_{K(p)} + \underbrace{V(x, y, z)}_{V(q)} \end{aligned}$$

Es.: $\mathcal{L}$ di una part. immerso in un campo E.M.:

$$\begin{aligned} \text{div } \vec{E} &= \frac{\rho}{\epsilon_0} & \text{div } \vec{B} &= 0 & \vec{B} &= \text{rot } \vec{A} & \vec{E} &= -\nabla \phi - \frac{\partial \vec{A}}{\partial t} \\ \text{rot } \vec{E} &= -\frac{\partial \vec{B}}{\partial t} & \text{rot } \vec{B} &= \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} & \text{GAUGE:} & & & \\ & & & & \vec{A} &\rightarrow \vec{A}' = \vec{A} + \nabla f & \\ & & & & \phi &\rightarrow \phi' = \phi - \frac{\partial f}{\partial t} \end{aligned}$$

$$\vec{A} = \begin{bmatrix} A_x(x, y, z, t) \\ A_y(x, y, z, t) \\ A_z(x, y, z, t) \end{bmatrix} \quad \phi = \phi(x, y, z, t)$$

$$\mathcal{L} = \underbrace{\frac{1}{2} m |\vec{v}|^2}_{E_K} - e \phi + \underbrace{e \vec{v} \cdot \vec{A}}_{\vec{F}_{el} \cdot \vec{r} = \hbar e \vec{v} \cdot \vec{A}}$$

$$\Rightarrow \mathcal{L} = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2) - e \phi + e v_x A_x + e v_y A_y + e v_z A_z$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = \frac{\partial \mathcal{L}}{\partial q_i} \quad q_i = x, \quad \dot{q}_i = \dot{x} = v_x$$

$$\Rightarrow \mathcal{L}_x = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial v_x} \right) = \frac{\partial \mathcal{L}}{\partial x}$$

(1) (2)

$$(1) \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial v_x} \right) = \frac{d}{dt} (m v_x + e A_x) = m \dot{v}_x + e \frac{dA_x}{dt} =$$

$$= m \dot{v}_x + e \left(\frac{\partial A_x}{\partial x} v_x + \frac{\partial A_x}{\partial y} v_y + \frac{\partial A_x}{\partial z} v_z + \frac{\partial A_x}{\partial t} \right)$$

$$(2) \frac{\partial \mathcal{L}}{\partial x} = -e \frac{\partial \phi}{\partial x} + e \left(\frac{\partial A_x}{\partial x} v_x + \frac{\partial A_y}{\partial x} v_y + \frac{\partial A_z}{\partial x} v_z \right)$$

$$m \dot{v}_x + e \left(\frac{\partial A_x}{\partial x} v_x + \frac{\partial A_x}{\partial y} v_y + \frac{\partial A_x}{\partial z} v_z + \frac{\partial A_x}{\partial t} \right) = -e \frac{\partial \phi}{\partial x} + e \left(\frac{\partial A_x}{\partial x} v_x + \frac{\partial A_y}{\partial x} v_y + \frac{\partial A_z}{\partial x} v_z \right)$$

$$\Rightarrow m \dot{v}_x = -e \left(\cancel{\frac{\partial A_x}{\partial x} v_x} + \frac{\partial A_x}{\partial y} v_y + \frac{\partial A_x}{\partial z} v_z + \frac{\partial A_x}{\partial t} \right) - e \frac{\partial \phi}{\partial x} + e \left(\cancel{\frac{\partial A_x}{\partial x} v_x} + \frac{\partial A_y}{\partial x} v_y + \frac{\partial A_z}{\partial x} v_z \right)$$

$$\Rightarrow m \dot{v}_x = -e \underbrace{\left[\frac{\partial \phi}{\partial x} + \frac{\partial A_x}{\partial t} \right]}_{-e E_x} + e \left[v_y \underbrace{\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)}_{(\text{rot } \vec{A})_z = B_z} + v_z \underbrace{\left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right)}_{-(\text{rot } \vec{A})_y = -B_y} \right]$$

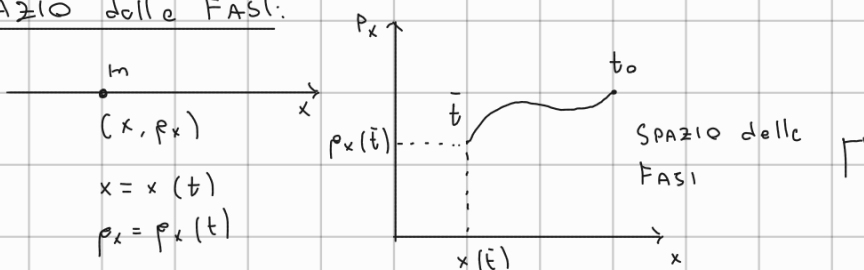
$$= -e E_x + e (v_y B_z - v_z B_y) = -e E_x + e (\vec{v} \wedge \vec{B})_x$$

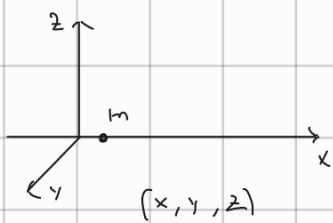
$$\Rightarrow m \dot{v}_x = -e E_x + e (\vec{v} \wedge \vec{B})_x \Rightarrow 3D \Rightarrow m \vec{\dot{v}} = -e \vec{E} + e \vec{v} \wedge \vec{B} \quad (\text{NEWTON con FORZA di LORENTZ})$$

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} + e A_x, \quad \vec{p} = m \vec{v} + e \vec{A} \quad m \vec{v} = \vec{p} - e \vec{A}$$

$$H = \sum_{i=x,y,z} p_i \dot{q}_i - \mathcal{L}, \quad H = K + V \quad \Rightarrow \quad \boxed{H = \frac{(\vec{p} - e \vec{A})^2}{2m} + e \phi} \quad \begin{array}{l} \text{e.h. pot. elettrost} \\ \downarrow \text{solo } \vec{v} \end{array}$$

SPAZIO delle FASI:





(p_x, p_y, p_z)

$[p]$

lo spazio 6D



$$d\Gamma = dq dp$$

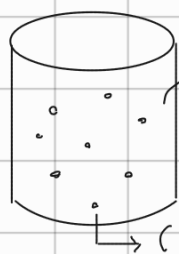
$[q]$ assi

N part. (3D) $\Rightarrow$ 6N

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \Rightarrow p_i \times \dot{q}_i = [\mathcal{L}] \text{ (dim. di una Lagrang., energia)}$$

$$= E \cdot t \rightarrow \text{MOMENTO ANGOLARE} \rightarrow \text{AZIONE}$$

FONDAMENTI di MECCANICA STATISTICA: ENSEMBLE



GAS PERF. MONAT.

1 mole $N_A = 10^{24}$ p, V, T (gas)

(x, y, z)

(p_x, p_y, p_z)

$\approx 10^{24}$ Eq. di HAMILTON

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = - \frac{\partial H}{\partial q_i} \quad (q_i(0), p_i(0))$$

$$G = G(x_1, \dots, x_N, p_1, \dots, p_N) = G(q, p)$$

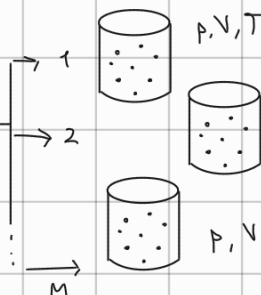
$$\bar{G} = \frac{1}{T} \int_0^T G(q(t), p(t)) dt$$

MEDIA TEMPORALE della grandezza G

10^{24}

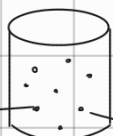
$\begin{cases} q_i(0) \\ p_i(0) \end{cases}$ NON FATTIBILE

Supp. di costruire M repliche del barattolo



COND. INIZ. DIVERSE

$M=3$



$t=t_1$

$q_1^{(1)}, p_1^{(1)}$

$q_1^{(3)}, p_1^{(3)}$

$q_N^{(3)}, p_N^{(3)}$

$[p]$

$t=t_1$

$t=t_2$

$[q]$

$[p]$

$[q]$

$[p]$

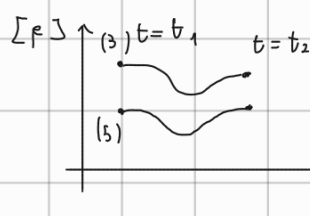
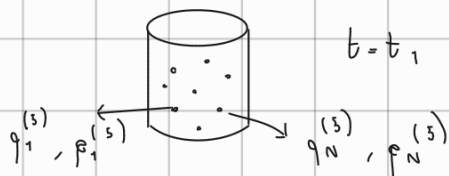
$[q]$

$$G(q_1^{(3)}, p_1^{(3)}, \dots, q_N^{(3)}, p_N^{(3)}) = G_3$$

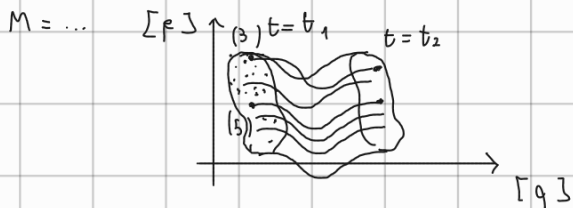
ENSEMBLE

(insieme M repliche)

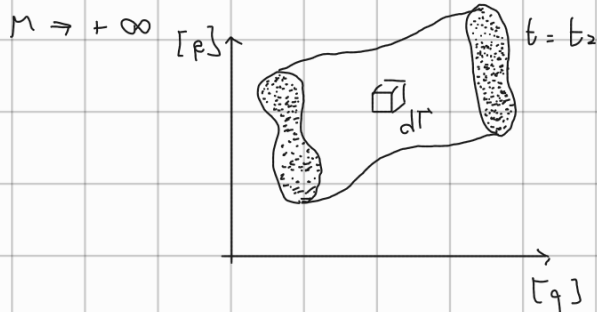
$$M=3$$



$$G(q_1^{(s)}, p_1^{(s)}, \dots, q_n^{(s)}, p_n^{(s)}) = G_s$$



$$\langle G \rangle = \frac{1}{M} \sum_i G_i \quad \text{media pesata sull'ensemble}$$



$$P(q, p) = \frac{\text{\# sistemi che hanno } (q, p)}{\text{Vol. in } \Gamma}$$

per tale valore

$$dM(q, p) = P(q, p) d\Gamma$$

$$\langle G \rangle = \frac{\int G(q, p) P(q, p) dq dp}{\int P(q, p) dq dp}$$

$$\Leftrightarrow \langle G \rangle = \frac{1}{M} \sum_i G_i$$

$$P(q, p) = \frac{\text{\# sist. che hanno } (q, p)}{\text{Vol } \Gamma} \Rightarrow \int P(q, p) d\Gamma = 1 \Rightarrow \text{DENSITÀ di PROBABILITÀ}$$

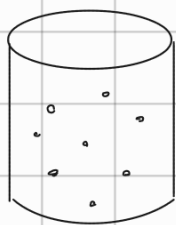
densità di prob. classica

$$\langle G \rangle = \int G(q, p) P(q, p) d\Gamma$$

$$\text{IPOTESI ERGOTICA} \Rightarrow \bar{G} = \langle G \rangle$$

Il punto fond. della mecc. stat. è trovare $P(q, p) = ?$

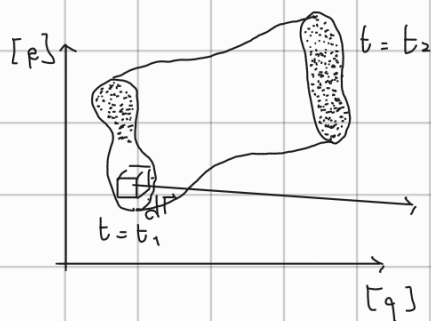
FONDAMENTI di MECCANICA STATISTICA: Ensemble



p, V, T

M repliche $\Rightarrow$ Ensemble

$M \Rightarrow +\infty$



$$d\Gamma = dq dp$$

$$G(q_i, p_i) = \rho(q_i, p_i) = \frac{\# \text{ Repliche con } (q_i, p_i)}{\text{Vol. in } \Gamma} = G_i = G(q_i, p_i)$$

$$\langle G \rangle = \frac{\sum G_i \times \# \text{ Repliche}}{\# \text{ Repliche tot}} = \frac{\int G(q, p) \times \overbrace{\rho(q, p) dq dp}^{\# \text{ repliche}}}{\int \rho(q, p) dq dp} \quad \begin{array}{l} \text{MEDIA PESATA} \\ \text{SULL' ENSEMBLE} \end{array}$$

$$\int \rho(q, p) dq dp = 1 \Rightarrow \text{densità di prob.}$$

$$\langle G \rangle = \int G(q, p) \rho(q, p) dq dp$$

IPOTESI ERGODICA:

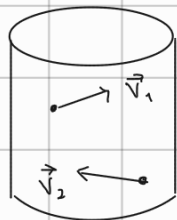
$$\bar{G} = \langle G \rangle$$

TEOREMA di LIOUVILLE:

$$\frac{d}{dt} [\rho(q, p)] = 0$$

STAZIONARIA

$$\rho(q, p) \Rightarrow \rho(E) = \rho[E(q, p)]$$



$$\textcircled{1} \quad \bar{v} = \frac{1}{T} \int_0^T v(x_1, y_1, z_1, p_{2x}, p_{2y}, p_{2z}) dt$$

$$\textcircled{2} \quad M \text{ REPLICHE} \quad \langle v \rangle = \frac{\int v(q, p) \rho(q, p) dq dp}{\int \rho(q, p) dq dp}$$

$$\bar{v} = \langle v \rangle$$

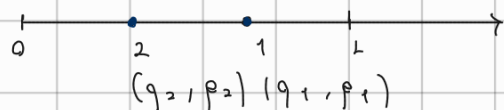
• ENSEMBLE MICROCANONICO:

SISTEMI ISOLATI $\rightarrow E$ si conserva

$$E = E_0 + \delta E$$

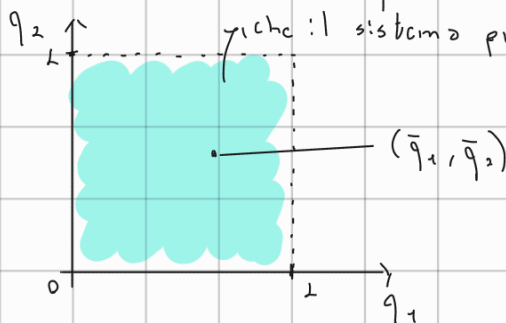
$$p(E) = \begin{cases} \text{cost} & \text{PER } E \equiv E_0 \\ 0 & \text{altrove} \end{cases}$$

E_s

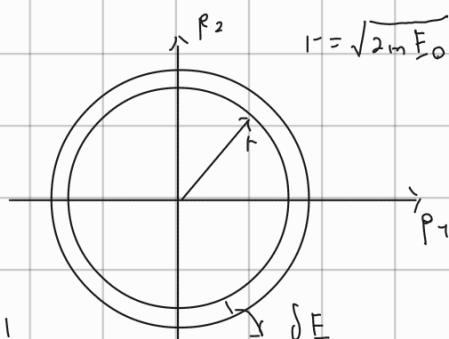


COORDINATE ACCESSIBILI:

tutte le possibili combinazioni
che il sistema può avere (ensemble)



$$\frac{p_1^2}{2m} + \frac{p_2^2}{2m} = E_0 \quad p_1^2 + p_2^2 = 2m E_0$$

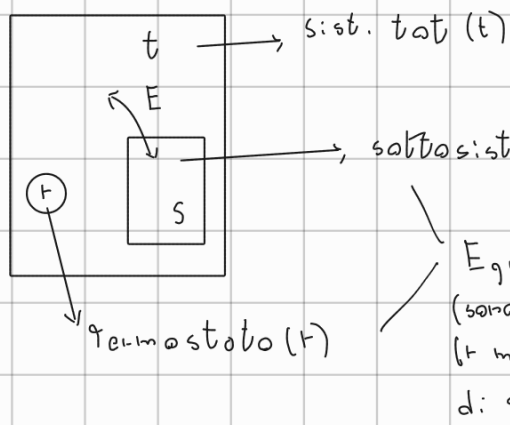


$\Gamma(q_1, q_2, p_1, p_2)$ (cambino il quadrato
e la circonf. e ottengo il
vol. in 4D che replica tutti i
sist. possibili)

Tutti i sistemi hanno la stessa E_0 , quindi la prob. di avere una replica
con val. $(\bar{q}_1, \bar{q}_2, \bar{p}_1, \bar{p}_2)$ è la stessa per tutti i sist., quindi $p(E) = \text{cost}$

• ENSEMBLE CANONICO:

SISTEMI in eq. TERMICO



NON ISOLATO $\Rightarrow$ SCAMBIA E col
THERMOSTATO

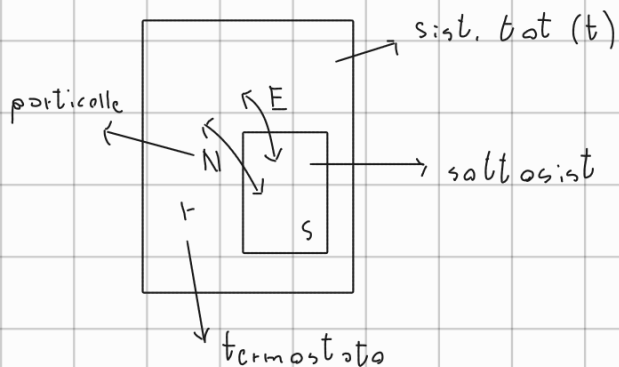
Equilib. termico
(sono in contatto)
T mantiene la temp.
di S)

le M repliche di S hanno una E diversa tra loro

Si: DEFINISCE $\langle E \rangle$ e ΔE (incertezza)

Si dimostra che $\Rightarrow \frac{\Delta E}{\langle E \rangle} \propto \frac{1}{\sqrt{N}} \xrightarrow{N \rightarrow +\infty} 0$

• ENSEMBLE GRANCANONICO:



Ogni replica del sottosist. ha una diversa

E e un diverso N

$\langle E \rangle$ e ΔE + $\langle N \rangle$ e ΔN

Riprendo ENS. CANONICO: Eq. term.

$$p(E) = C e^{-\beta E} \quad C = \text{cost}, \quad \beta = \frac{1}{k_B T} \quad k_B = 1,38 \cdot 10^{-23} \text{ J/K}$$

$$= C e^{-\frac{E}{k_B T}}$$

$\Rightarrow$ DISTRIBUZIONE DI BOLZMANN

① $T = 300 \text{ K}$

$$\Rightarrow k_B T = 9,1 \cdot 10^{-21} \text{ J}$$

$$= 2,2 \cdot 10^{-2} \text{ eV}$$

$$= 22 \text{ meV}$$

INTRODUKO: $1 \text{ eV} = 1,6 \cdot 10^{-19} \text{ J}$ (En. cinetica di un elett. sotto diff. pot. di un volt)

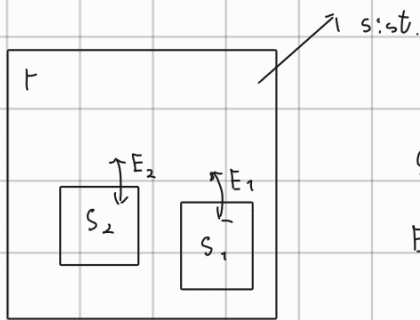
$$p = C e^{-\beta E} \Rightarrow \int p(q, p) dq dp = 1 = C \int e^{-\beta E} dq dp \Rightarrow C = \frac{1}{\int e^{-\beta E} dq dp}$$

$$Z = \int e^{-\beta E} dq dp \quad \text{FUNZIONE PARTIZIONE}$$

(se C ha questo val. p è una dens. di prob.)

$$p(E) = \frac{e^{-\beta E}}{Z} \Rightarrow \text{NORMALIZZATA} \Rightarrow \text{dens. di prob.}$$

$$\langle U \rangle = \frac{\int U(q, p) e^{-\beta E} dq dp}{Z}$$



S_1 e S_2 distinti e indipend.

$$E_{1+2} = E_1 + E_2$$

$$Z_{1+2} = \int e^{-\beta E_{1+2}} dq_1 dp_1 dq_2 dp_2 =$$

$$= \int e^{-\beta(E_1+E_2)} dq_1 dp_1 dq_2 dp_2 = Z_1 \cdot Z_2$$

N sottosist. $\Rightarrow Z_N = \prod_{j=1}^N Z_j$

$$\langle E \rangle = U = \frac{\int E(q, p) e^{-\beta E} dq dp}{Z} \quad ; \quad \frac{\partial}{\partial \beta} \ln(Z) = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = \frac{1}{Z} \frac{\partial}{\partial \beta} \int e^{-\beta E} dq dp =$$

$$= \frac{1}{Z} \int \frac{\partial}{\partial \beta} (e^{-\beta E}) dq dp = -\frac{1}{Z} \int E e^{-\beta E} dq dp$$

$$\Rightarrow \langle E \rangle = U = -\frac{\partial}{\partial \beta} \ln Z$$

TEOREMA di EQUIPARTIZIONE dell'ENERGIA: (per sist. a eq. termico)

Es.: una delle variabili q_i del cos

$$E = \overset{\uparrow}{2x^2} + f(q, p), \quad q \neq x \quad \Rightarrow \quad \langle E \rangle = \underbrace{\langle 2x^2 \rangle}_{\uparrow \text{trascurato}} + \underbrace{\langle f(q, p) \rangle}_{\uparrow \text{trascurato}}$$

$$\langle 2x^2 \rangle = \frac{\int 2x^2 e^{-\beta E} dq dp}{Z} = \frac{\int 2x^2 e^{-\beta(2x^2+f)} dq dp}{\int e^{-\beta(2x^2+f)} dq dp} = \frac{\int 2x^2 e^{-\beta 2x^2} dx \int e^{-\beta f} (dq dp)_{q \neq x}}{\int e^{-\beta 2x^2} dx \int e^{-\beta f} (dq dp)_{q \neq x}}$$

INTEGR. di una Gaussiana

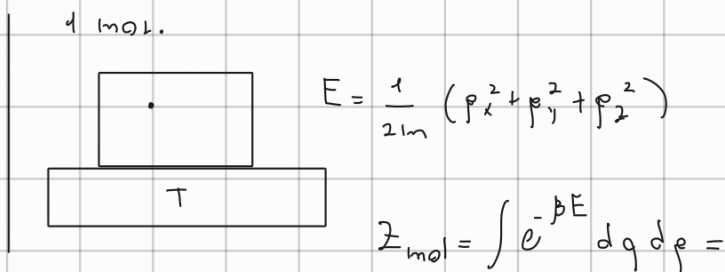
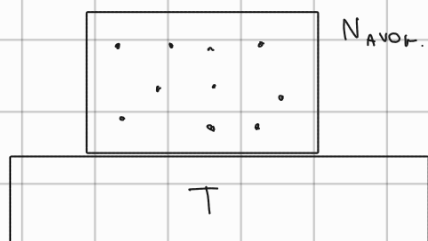
$$= \frac{\int 2x^2 e^{-\beta 2x^2} dx}{\int e^{-\beta 2x^2} dx} = -\frac{\partial}{\partial \beta} \ln \left[\int e^{-\beta 2x^2} dx \right] = -\frac{\partial}{\partial \beta} \ln \left[\sqrt{\frac{\pi}{\beta 2}} \right] = -\frac{1}{2} \frac{\partial}{\partial \beta} [\ln \pi - \ln \beta - \ln 2] =$$

$$= \frac{1}{2\beta} = \frac{1}{2} k_B T \quad \Rightarrow$$

$$\langle 2x^2 \rangle \Rightarrow \frac{1}{2} k_B T$$

$$E = \underbrace{ax^2}_{\text{pot.}} + \underbrace{bp^2}_{\text{cin.}} + cx_1 + dp_1 \Rightarrow \langle E \rangle = \underbrace{\frac{1}{2} k_B T}_{\text{pot.}} + \underbrace{\frac{1}{2} k_B T}_{\text{cin.}} + \langle cx_1 + dp_1 \rangle$$

APPLICAZIONE: GAS PERFETTO MONOATOMICO:



$$= \int e^{-\beta \left[\frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) \right]} dx dy dz dp_x dp_y dp_z$$

$$= V \cdot \left(\int e^{-\beta \frac{p^2}{2m}} dp \right)^3 = V \cdot (2\pi m k_B T)^{3/2}$$

Tutte le altre molecole sono N_A sottosistemi non interagenti

$$\Rightarrow Z_{gas} = \prod_{i=1}^{N_A} Z_{mol,i} = \boxed{V^{N_A} (2\pi m k_B T)^{\frac{3N_A}{2}}}$$

$$\langle E \rangle = - \frac{\partial}{\partial \beta} \ln(Z_{gas}) = U_{gas} = - \frac{\partial}{\partial \beta} \ln \left[V^{N_A} \left(\frac{2\pi m}{\beta} \right)^{\frac{3N_A}{2}} \right] = - \frac{\partial}{\partial \beta} \ln \left[\frac{V^{2/3} 2\pi m}{\beta} \right]^{\frac{3N_A}{2}}$$

$$= - \frac{3N_A}{2} \frac{\partial}{\partial \beta} \ln \left[\frac{V^{2/3} 2\pi m}{\beta} \right] = - \frac{3N_A}{2} \left(\frac{1}{\frac{V^{2/3} 2\pi m}{\beta}} \right) \left(- \frac{V^{2/3} 2\pi m}{\beta^2} \right) =$$

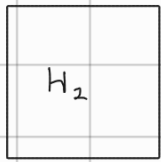
$$= \frac{3N_A}{2} \frac{1}{\beta} = \frac{3}{2} \overbrace{N_A}^R k_B T = \frac{3}{2} RT$$

$$\bullet E_{1mol} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} \Rightarrow \langle E_{1mol} \rangle = \frac{1}{2} k_B T + \frac{1}{2} k_B T + \frac{1}{2} k_B T = \frac{3}{2} k_B T$$

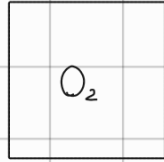
$$\Rightarrow \langle E_{gas} \rangle = \langle E_{1mol} \rangle N_A = \frac{3}{2} N_A k_B T = \frac{3}{2} RT$$

$$C_V = \left(\frac{\partial U}{\partial T} \right) = \frac{3}{2} R$$

ATOMI e PARTICELLE: metodi speriment. e classici



stessa
p, V, T



$$\frac{m_{H_2}}{m_{O_2}} = \frac{1}{16}$$

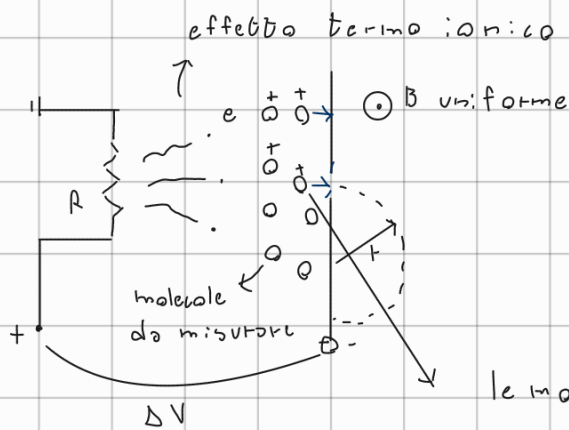
⇒ Peso Atomico (Peso molecolare) H → 1 H₂ → 2 , O → 16 O₂ → 32

1 mol sostanza	H ₂	O ₂	Ar	H ₂ O
Peso Atom.	H → 1 H ₂ → 2	O → 16 O ₂ → 32	Ar → 40	H → 1 O → 16 H ₂ O → 18
MASSA Atomica [g]	2g	32g	40g	18g

$$1 \text{ u.m.z.} = m_H \approx m_{\text{proton}} = m_{\text{neutrone}}$$

$$= \frac{1}{2} \frac{(1 \text{ mol H}_2)}{N_A} = 1,67 \cdot 10^{-24} \text{ g}$$

• Misura della massa: Spettrometro di massa



$$\frac{1}{2} m v^2 = q \Delta V \Rightarrow v = \sqrt{\frac{2 q \Delta V}{m}}$$

$$q v B = m \frac{v^2}{r} \Rightarrow r = \frac{m v}{q B}$$

le molecole si ionizzano

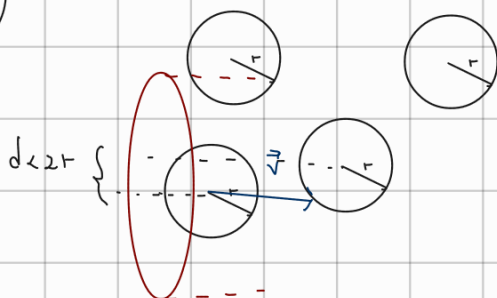
$$\Rightarrow r^2 = \frac{2 m \Delta V}{q B^2} \Rightarrow \frac{q}{m} = \frac{2 \Delta V}{r^2 B^2}$$

NOTO
↓
misura

STIMARE RAGGIO Atomico:



① Libero cammino medio: l_m = distanza percorsa da un atomo tra 2 urti



n → densità

$$\left. \begin{aligned} V_{cyl} &= v \Delta t \\ S_{cyl} &= \pi (2r)^2 \end{aligned} \right\} V_{cyl} = n \pi r^2 v \Delta t$$

$$N_{\text{stom}} \text{ nel cilindro} = n \cdot V_{\text{cil}} = n \cdot 4\pi r^2 v \Delta t \quad \ell_m = \frac{\text{distanza percorsa in } \Delta t}{\# \text{ urti in } \Delta t}$$

$$= \frac{v \Delta t}{4\pi r^2 v \Delta t} = \frac{1}{4\pi r^2} \quad (\text{se tutti gli stomi si muovono mult. per } \sqrt{2})$$

Viscosimetri misurano ℓ_m (Es. in aria $\ell_m = 10^{-7} \text{ m}$)

② Eq. di stato dei GAS REALI: 1 mol

$$\left(p + \frac{a}{V^2}\right) (V - b) = RT$$

$$b = \text{covolume} = n \cdot \overbrace{\frac{4}{3} \pi r^3}^{\text{Vol. stomo}} \cdot N_A$$

Fisso V

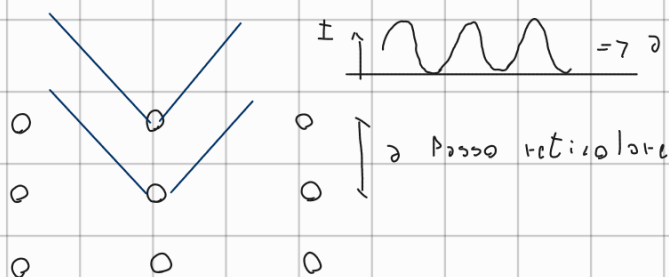
T	P
~	~
~	~
~	~
~	~
~	~

$\Rightarrow$ Trovo P in funz. di T

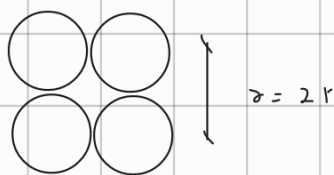
$\Rightarrow$ Fitto: dati trovando a e b

$\parallel$ ricavo r

③ DIFFRAZ. di BRAGG:



In realtà



N_e

$$r_{Ne} \begin{cases} \rightarrow 1,18 \cdot 10^{-10} \text{ m} & (\ell_m) \\ \rightarrow 2,1 & \text{"} & (\text{Eq. gas reali}) \\ \rightarrow 1,6 & \text{"} & (\text{Bragg}) \end{cases}$$

H

$$Z = 1 \quad r_H = 0,53 \text{ \AA} (10^{-10} \text{ m})$$

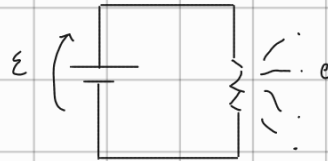
U

$$Z = 92 \quad r_U = 1,4 \text{ \AA}$$

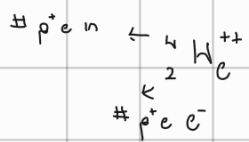
$\downarrow$ Uranio

• ELETRONI: $m_e = \frac{1}{1836} m_p = 9,1 \cdot 10^{-31} \text{ kg}$ PARTICELLE ELEMENTARI

$q_e = -1,6 \cdot 10^{-19} \text{ C}$ S: generano per eff. termoionico



• PARTICELLE α : $m_\alpha = 4 u.m.a \rightarrow$ Nuclei di He

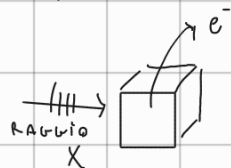


S: generano da decadimenti radioattivi $\Rightarrow E_k = 1-10 \text{ MeV}$

$\ell_m = 10 \text{ cm}$, $v \approx 10^7 \text{ m/s}$

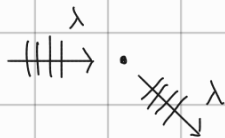
• RAGGI X: LUCE $\lambda \approx 1 \text{ \AA}$

$\rightarrow$ ASSORBIMENTO:

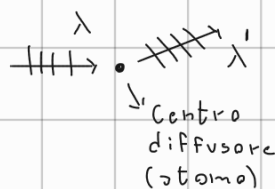


EFFETTO FOTOELETTRICO

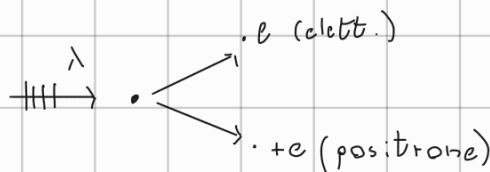
$\rightarrow$ DIFFUSIONE ELASTICA: (scattering Rayleigh)



$\rightarrow$ DIFFUSIONE ANELASTICA: (scattering Compton)

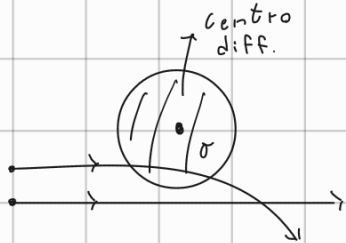
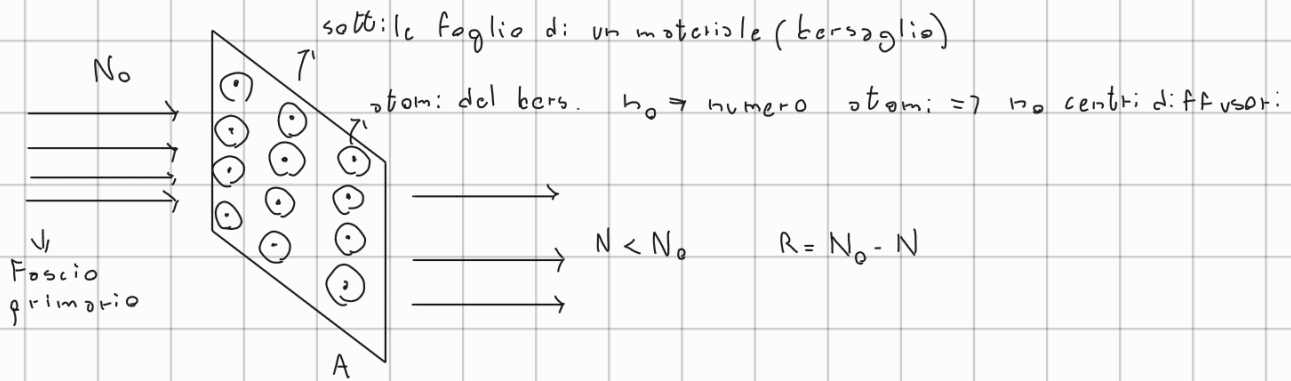


$\rightarrow$ CREAZIONE di COPPIE:



solo se $\Rightarrow E > 2 m_e c^2$

SEZIONE d'URTO: Utilizzo SONDE (es. fascio di elett. / raggi X)



$\sigma \Rightarrow$ sezione d'urto [m^2]

Prob. di un evento di rimozione dal fascio

$$P = \frac{n_0 \cdot \sigma}{A} \quad \text{se } n_0 \sigma = A \Rightarrow \text{non passa nulla}$$

$$P = n \sigma, \quad R = P \cdot N_0 = n \sigma N_0$$

$\downarrow$
 $\frac{n_0}{A}$

$$\sigma_{TOT} = \sigma_{FOTOEL} + \sigma_{COMP.} + \sigma_{RAY.}$$

$$\sigma_{TOT} = \int \sigma$$

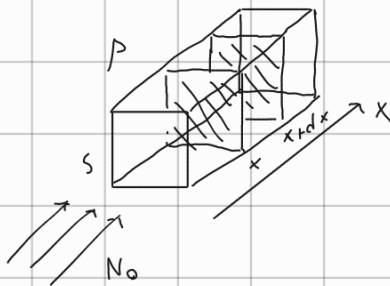
$$\rho = \frac{Kg}{m^3}$$

Atomi

Vol

Cent. d'ff

Vol



P ev. di rim.

$$-dN = [N(x+dx) - N(x)] = \sigma n N(x)$$

$$\Rightarrow n = p dx$$

$$\Rightarrow -\sigma p N(x) dx = dN$$

$$N(x) = N_0 e^{-\sigma p x}$$

$$N(x) = N_0 e^{-x/\Lambda}$$

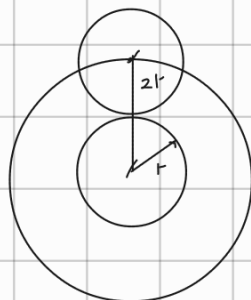
$\Lambda = \frac{1}{\sigma p}$ Libero cammino medio

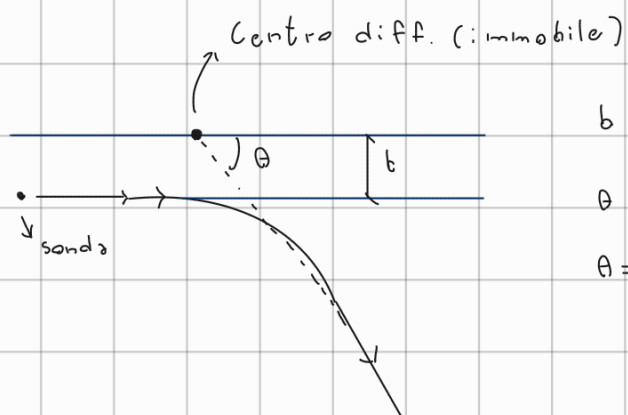
$$l_m = \frac{1}{4\pi r^2 n} = \frac{1}{\sigma p}$$

Atomi

Vol

$\Rightarrow \sigma = 4\pi r^2$

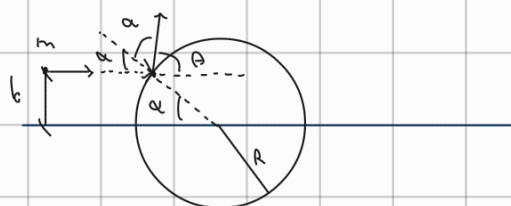




b = parametro d'impatto

θ = angolo di diffusione

$$\theta = \theta(b)$$



$$\theta = \theta(b)$$

$$b = R \sin \alpha \quad 2\alpha + \theta = \pi$$

$$\Rightarrow \alpha = \frac{\pi}{2} - \frac{\theta}{2} \Rightarrow b = R \sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right) = R \cos\left(\frac{\theta}{2}\right)$$

$$\theta = \begin{cases} 2 \arccos\left(\frac{b}{R}\right) & b \leq R \\ 0 & \text{altrimenti} \end{cases}$$